

DEJAN GRUBISIC

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[Website](#) | [Google Scholar](#) | [LinkedIn](#)

EDUCATION

Doctor of Philosophy <i>High Performance Computing and Machine Learning</i> Rice University GPA: 3.56	Aug. 2019 – May 2024 Houston, TX
Master of Science <i>Big Data Architectures</i> University of Novi Sad GPA: 4.00	Aug. 2018 – May 2019 Novi Sad, Serbia
Bachelor of Science <i>Electrical and Computer Engineering</i> University of Novi Sad GPA: 3.96	Aug. 2014 – May 2018 Novi Sad, Serbia

For all passed courses check here [All Courses](#)

WORK EXPERIENCE

Axiom Math Member of Technical Staff Developing the best AI model for mathematics	Jun. 2025 – Present Palo Alto, CA
Meta AI AI Research Scientist - Worked on modern recommendation systems and Foundational AI Research (FAIR) - Contributed to Meta's LLM Compiler, a suite of foundation models for compiler optimization	Jul. 2024 – Jun. 2025 Menlo Park, CA
Rice University Graduate student - Developing GPU support for large-scale profiler - HPCToolkit - Creating analysis techniques for GPU accelerated applications - Developing profiler guided optimizations based on reinforcement learning - Compiler2	Aug. 2019 – May 2024 Houston, TX
Meta AI Machine Learning Research Intern - Finetuning Llama2-based model on LLVM IR programs to solve phase-ordering problem - Analyzing LLM's capability in applying 60 LLVM optimizations	May 2023 – Dec. 2023 Menlo Park, CA
Meta AI Machine Learning Research Intern - Building reinforcement learning framework - LoopTune for optimizing tensor operations - Building and tuning cost and policy deep learning models	May 2022 – Dec. 2022 New York City, NY
Berkeley Lab High Performance Computing Research Intern Profiling and analysis of power consumption on multi node GPU applications	Jun. 2021 – Sep. 2021 Berkeley, California
Rice University Research intern Optimization of MADNESS (Multiresolution Adaptive Numerical Environment for Scientific Simulation) in Intel CnC workframe	Jun. 2018 – Sep. 2018 Houston, TX
Institute for High Performance Microelectronics Hardware Engineer Intern - Profiling and Analysis of FFT implementation on Xtensa Platform in C - Dhrystone Benchmark for FFT and theoretical analysis window functions	Jun. 2017 – Sep. 2017 Frankfurt O, Germany

PUBLICATIONS

ABC implies that Ramanujan's tau function misses almost all primes Indagationes Mathematicae (also on arXiv)	Mar. 2026
Almost all primes are partially regular Preprint available on arXiv	Feb. 2026
Dead ends in square-free digit walks Preprint available on arXiv	Feb. 2026
Fel's Conjecture on Syzygies of Numerical Semigroups Preprint available on arXiv	Feb. 2026
LLM Compiler: Foundation Language Models for Compiler Optimization ACM SIGPLAN Int. Conf. on Compiler Construction (CC '25)	Mar. 2025
Priority Sampling of Large Language Models for Compilers Workshop on Machine Learning and Systems (MLSys)	Feb. 2024
Compiler Generated Feedback for Large Language Models Preprint available on arXiv	Mar. 2024
Large Language Models for Compiler Optimization Preprint available on Arxiv	Sep. 2023
LoopTune: Optimizing Tensor Programs with Reinforcement Learning Preprint available on Arxiv	Sep. 2023
LoopStack: ML-friendly ML Compiler Stack Neural Information Processing Systems	Nov. 2022
Measurement and Analysis of GPU-Accelerated OpenCL Computations on Intel GPUs International Workshop on Programming and Performance Visualization Tools	Nov. 2021
An Automated Tool for Analysis and Tuning of GPU-accelerated Code in HPC Applications IEEE Transactions on Parallel and Distributed Systems	Feb. 2021
Measurement and Analysis of GPU-accelerated Applications with HpcToolkit Parallel Computing Journal	Nov. 2020
Finding multi-source shortest path in dynamic large-scale graph in Lambda architecture Faculty of Technical Sciences in Novi Sad Journal	May 2019

RESEARCH AND PROJECTS

Statistical Machine Learning

Kaggle challenge: NBME - Score Clinical Patient Notes

- Fine tuning Bert model to predict features from the patient notes (Python)

Artificial Intelligence

Pacwar: Finding the strongest gene by using genetic algorithms

- Hill climbing, genetic algorithm, finding local hills with K-means, defining scoring function and other tricks in (Python)

Multiprocessing

Lock free concurrent skip list

- Implemented using compare-and-swap atomic primitives and OpenMP (C)

Parallel Computing

Parallel algorithms in various technologies

- Parallel exploratory search of game Othello using Cilk (C++)
- LU decomposition using OpenMP (C++)
- 2.5 matrix multiplication using OpenMPI (C++)

- Bitonic sort using Cuda (CudaC++)

Compiler Construction 2

Compiler for DEMOgram language

- Compiler for custom language implemented in Flex/Bison framework (C)

Compiler Construction 1

Compiler for ILOC language

- Implementating scanner, parser, registrar allocator and instruction schedulers implemented (C++)

Master Thesis

Finding multi-source shortest path in dynamic large-scale graph, based on Lambda architecture

- Implementing batch and real-time processing algorithms (pySpark)
- Implementing storage in HDFS and communication in Kafka framework
- Creating web based application (Python Dash)
- Using Docker for containerization

Data Science

Movie profit prediction

- Predicting audience interest in movie by using logistic regression, k-nearest neighbor, support vector machines
- Predicting weekend profit for a movie by using linear regression, neural network, SVM and NaiveBayes

Bachelor Thesis

FPGA design of hardware core for acceleration of chess engine

- Analysis and design of software-hardware system using ESL methodology (SystemC)
- Implementing board evaluation module in VHDL
- Verification of design using UML methodology (SystemVerilog)

Applied Electronics

Metal Detector

- End-to-end implementation of circuit, PCB and packing with AltiumDesigner

COMMUNITY AND LEADERSHIP

Southeast Europe AI Summit

Jan. 2026 – Present

Founder

Remote

- Founded a summit dedicated to growing the AI ecosystem and connecting researchers, builders, and companies across Southeast Europe

Serbian AI Society

Jun. 2023 – Present

Ecosystem Team Lead

Novi Sad, Serbia

- Leading ecosystem-building efforts to foster the growth of the Serbian AI community

SKILLS

Programming Languages: Python, C/C++, CudaC, Lean, Java, Bash, VHDL

Technologies: PyTorch, OpenMP, MPI, CUDA, Docker, Slurm, Spark, Linux

Featured Skills: Data Preparation, Fine-Tuning, LLMs, Reinforcement Learning, Compiler Construction, Parallel Computing